

■ STATIC ELECTRICITY ■

DANCING CEREAL

Detailed Lesson Plan · Planrs Academy Science Lab · Session 1

Duration	Session	Type	Subject	Depth
30 Minutes	Session 1	Hands-on Lab	Science / EVS	Beginner

1. LESSON OVERVIEW

What this session is about and why it matters

This session introduces learners to the magical, invisible world of **STATIC ELECTRICITY** through one of science's most dramatic demonstrations — making cereal dance without touching it! By rubbing a plastic comb (kanghi) or balloon against dry hair or wool, learners build up static charge, then watch in wonder as puffed rice (murmura) leaps, jumps, and "dances" toward the charged object. The session is built entirely around the **predict-test-observe-conclude** cycle, with Indian everyday objects making every concept instantly familiar and achievable.

■ The Big Idea: **Rub → Build Charge → Invisible Force → Attracts Light Things!** Just like a kanghi attracting hair every morning — static electricity is already in every Indian home!

Prior Knowledge Assumed

- ◆ Awareness of things that rub together
- ◆ General knowledge of bicycles and daily objects
- ◆ No prior electricity knowledge required
- ◆ Fine motor skills sufficient to rub a comb on hair

Forward Curriculum Links

- ◆ Simple circuits and current electricity (Class 4–5)
- ◆ Conductors and insulators (Class 6, NCERT)
- ◆ Electric charges and fields (Class 12)
- ◆ NEP 2020: observe, question, hypothesise competency

2. LEARNING OBJECTIVES

By the end of this session, learners will be able to:

LO	Learning Objective	Bloom's Level	Measured By
LO 1	Name static electricity as an invisible force created by rubbing two things together	Remember	Exit ticket / vocabulary recall
LO 2	Explain why a charged comb or balloon attracts murmura and small light objects	Understand	Discussion questions
LO 3	Predict which objects will be attracted before testing and compare result to prediction	Apply	Observation worksheet
LO 4	Give at least one real-life Indian example of static electricity from daily experience	Apply / Analyse	Think and Share discussion
LO 5	Participate in the dancing cereal investigation and record observations on at least 3 materials	Analyse	Worksheet completion

■ Success Indicator: A learner has achieved all objectives when they can say — "Static electricity is made when we rub things together. The charge is invisible but it pulls light things toward it — like magic! My kanghi pulled the murmura because of static electricity!"

3. KEY CONCEPTS WITH INDIAN EXAMPLES

Core science ideas for this session — anchored in everyday Indian life

Concept 1 — What is Static Electricity?

Static electricity is an **invisible force that builds up on the surface of objects when they are rubbed together**. "Static" means staying in one place — unlike current electricity which flows through wires. When objects are rubbed, charged particles move from one surface to the other, creating an imbalance. This charge imbalance produces the force we call static electricity.

Indian Example	What is Happening	Static or Not?
Kanghi (comb) attracting hair on a dry morning	Plastic comb rubbed on dry hair builds charge; charge attracts hair	YES — Classic static!
Woollen sweater crackling when removed in Delhi winter	Wool and skin rub together, building charge; release causes tiny sparks	YES — Charge discharging!
Shock when touching metal school gate or bus railing	Charge built up on body discharges suddenly through the metal conductor	YES — Sudden discharge!
Murmura jumping toward charged plastic comb or balloon	Charged object attracts the ultra-light puffed rice via electrostatic force	YES — This session!

Indian Example	What is Happening	Static or Not?
Ball rolling down a ramp and stopping	This is friction and gravity — not static electricity	NO — Different force

Concept 2 — Why Does the Charged Object Attract?

When a plastic comb or balloon is rubbed on hair or wool, it gains extra charge. This charged object creates an **invisible pulling force** — similar to how a magnet pulls iron filings, but this is a completely different phenomenon. The charged object can attract very light, uncharged objects such as murmura, paper bits, salt grains, and dry tea leaves.

Objects the Charged Comb WILL Attract	Objects the Charged Comb Will NOT Attract
■ Murmura (puffed rice) — extremely light, jumps dramatically ■ Small torn newspaper or notebook paper bits ■ Salt grains (namak) — very small and light ■ Dry tea leaves (chai patti) — ultra-light ■ Thermocol / styrofoam pieces from packaging ■ Tiny wool thread snippings	■ Stone / pebble — too heavy for weak static force ■ Coin / key — too heavy AND metallic (conducts charge away) ■ Pencil — too heavy for the force ■ Rubber eraser — too heavy ■ Steel tiffin box — metal conducts charge away ■ Water — conducting liquid, drains the charge

Concept 3 — The Dancing Cereal Experiment

When a plastic comb or balloon is rubbed on dry hair or a woollen cloth, it builds up static charge. Held near a pile of murmura on a steel thali, the charged object ATTRACTS the puffed rice — making it leap, jump, and appear to dance! The steel thali works perfectly as a flat, stable surface that does not interfere with the static charge.

Item	Indian Name	Qty	Where to Get It	Purpose
Plastic comb	Kanghi	1	Any home or chemist shop	Main charge-builder (rub on hair)
Puffed rice	Murmura / Mamra	Small handful	Any kirana shop or home kitchen	Primary "dancing" material — very light
Balloon (inflated)	Gubbaara	1	Any local balloon-wala or bazar	Alternative charge-builder (rub on hair or wool)

Item	Indian Name	Qty	Where to Get It	Purpose
Steel plate or thali	Thaali	1	Every Indian kitchen	Surface to hold materials during test
Woollen cloth / monkey cap	Oonee topi / kapda	1	Any wardrobe (winter clothes)	For rubbing in absence of dry hair
Torn newspaper bits	Akhbaar ke tukde	10–15 pieces	Any newspaper	Test material — light paper bits
Salt grains	Namak	Small pinch	Every kitchen	Test material — tiny and light
A small stone / coin	Pathar / sikka	1	Everywhere!	Control object — heavy, will NOT jump
Observation worksheet	—	1 per learner	Print before session	Recording predictions and results

Concept 4 — Static Electricity in Indian Daily Life

Everyday Static Electricity (Indian Life)

- Kanghi attracting hair every morning
- Balloons sticking to walls at Diwali / birthday parties
- Plastic cling wrap / polythene clinging to a dabba
- Dust sticking to television screens and plastic surfaces
- Stickers sticking without glue — static attraction
- Synthetic dupatta clinging to skin on dry days

Surprising / Shocking Static Moments

- Shock from metal gate after sitting on plastic chair
- Sweater crackling in dark winter nights (Delhi / Shimla)
- Getting shocked when touching a friend after sliding on plastic
- Lightning during monsoon — nature's massive static discharge!
- Clothes clinging together from the washing machine
- Getting a shock from a car door handle in dry weather

■■ Lightning Connection: The **monsoon lightning** that Indian children witness is nature's own static electricity on a massive scale! Storm clouds rub together, build enormous charge, and discharge as lightning. The same invisible force — just a trillion times bigger!

4. SESSION OUTLINE WITH TIMING

Total Duration: 30 Minutes · 5 Phases

1 ■ Hook & Wonder

0:00–0:05 (5 mins)

Purpose: Ignite curiosity with a dramatic demonstration before any explanation.

The WOW Demo

Pick up a plastic kanghi. Comb dry hair 10 times quickly. Hold it above a small pile of murmura on a steel thali. The murmura LEAPS toward the comb! Let learners gasp — do NOT explain immediately. Let the wonder breathe.

Opening Questions

"Did you see that?! I did not touch the murmura — but it JUMPED! WHY did that happen?" Accept all responses enthusiastically. Write them on the board. "Has your hair ever stuck to your comb in the morning? Have you heard your sweater crackle? That is the SAME thing!"

Introduce the Word

"The magic has a name — STATIC ELECTRICITY! Say it with me: STA-TIC E-LEC-TRI-CI-TY! It is invisible. It is everywhere. And today we are going to make cereal DANCE with it!"

2 ■ Concept Introduction

0:05–0:12 (7 mins)

Purpose: Build core understanding — what static electricity is and why it causes attraction.

Step 1 · 2 min

Explain charge build-up: "Everything around us is made of tiny invisible particles. Some have a special power called CHARGE. When we RUB two things together, some charged particles move from one to the other. Now one thing has too many charges — and that extra charge is called STATIC ELECTRICITY!" Show: comb before and after rubbing (neutral vs. charged visual with glow effect).

Step 2 · 2 min

Why does it attract? "The charged comb creates an invisible pulling force — like an invisible magnet! It can pull very light things toward it." Indian connection: "Just like your kanghi attracts your hair on a dry morning! Or a rubbed balloon stuck to the wall at your friend's birthday party!"

Step 3 · 3 min

Static electricity in daily life — show 3 quick Indian examples on screen: (1) Kanghi attracting hair. (2) Sweater crackling in winter — "tiny sparks of static between wool and skin!" (3) Lightning in monsoon sky — "the BIGGEST static electricity discharge in all of nature!" Ask: "Can you think of one more place you have felt static electricity at home?"

3 ■ Hands-On Investigation

0:12–0:22 (10 mins)

Purpose: Learners experience static electricity directly through a joyful investigation.

Investigation: "Dancing Cereal — What Will the Charged Comb Attract?"

PREDICT (2 min)	Learners look at all test materials on the thali. They write/circle prediction on worksheet: "I think the charged comb WILL attract: _____ because _____. I think it WILL NOT attract: _____ because _____. Predictions are shared — celebrate all guesses, right or wrong.
CHARGE (1 min)	Rub the plastic comb on dry hair firmly, at least 10 times fast. Or rub balloon on hair / woollen monkey cap. Hold comb slightly above each test material. TEACHER DEMOS FIRST. Then each pair of learners tries. CRITICAL NOTE: Must be done in dry conditions. Humid or rainy days significantly reduce the static effect.
TEST EACH (4 min)	Hold charged comb 2–3 cm above each material. Observe and record: Murmura — leaps and dances (very dramatic!). Paper bits — flutter and some jump. Salt grains — some small ones jump. Stone/pebble — does NOT move (too heavy — key learning!). Record tick/cross on observation sheet for each.
THE CONTROL TEST	This step is CRITICAL: before rubbing, hold the plain (uncharged) comb above the murmura. Nothing happens! Then rub and try again — murmura dances! Ask: "Why did rubbing make a difference? What changed?" This proves it is the CHARGE from rubbing that causes attraction, not the comb itself.
OBSERVE & RECORD (3 min)	"Did it jump? YES / NO (circle one)." "I think it jumped / did not jump because _____. Learners draw what they saw. For early finishers: try dry atta, tea leaves, thermocol bits. For learners needing support: teacher holds comb; learner records observations.

■ Safety Note: No sharp objects. Balloons should not be over-inflated. If a learner gets a mild static tingle — normalise it: "That is the static electricity moving! That is exactly what we are studying!" Do NOT attempt near water or on humid / rainy days.

4

■ Discuss & Connect

0:22–0:27 (5 mins)

Purpose: Make sense of observations; connect to real life; deepen vocabulary.

Discussion Question	Expected Response / Teaching Note
Q1: Which objects jumped toward the comb? Which did NOT jump? Why?	Murmura, paper, salt jumped. Stone did not — too heavy for the weak static force. Guide: "The force is real but small — only very light objects feel it enough to move."
Q2: What happened when we held the UNCHARGED comb above the murmura?	Nothing happened! This proves that the CHARGE from rubbing is what causes attraction — not the comb itself. This is the most important scientific insight of the session.

Discussion Question	Expected Response / Teaching Note
Q3: Where have YOU felt or seen static electricity at home or on the street?	Prompt if needed: Kanghi and hair, crackling sweater, shock from gate or bus railing, balloons at Diwali party sticking to wall, TV screen collecting dust, polythene wrap clinging to food.

■ Indian Spotlight: "Every time you comb your hair with a plastic kanghi on a dry day — you are making static electricity! During monsoon, when you see lightning — that is static electricity on a MASSIVE scale! The dust on your TV screen? Static electricity is pulling dust particles to the plastic. That is why plastic collects more dust than metal!"

5

■ Wrap-Up & Recap

0:27–0:30 (3 mins)

3 Big Things We Learned Today:

1

Static electricity is made when we rub two things together

2

The invisible charge can attract (pull) very light objects toward it

3

We see static electricity every day — kanghi and hair, crackling sweaters, stickers sticking, and even monsoon LIGHTNING!

■ Exit Ticket — "STATIC! or NOT STATIC?": Show 3 images. Learners call out the answer. (1) Kanghi attracting hair → STATIC! (2) A ball rolling down a ramp → NOT STATIC! (3) Lightning in the monsoon sky → STATIC!

■ Take-Home Challenge — Static Electricity Detective: Try rubbing your kanghi on dry hair at home before bed. Hold it near small torn bits of paper. Did it work? Find ONE more thing at home that is attracted to the charged comb. Tell us tomorrow!

5. VOCABULARY GLOSSARY

Key words introduced in this session — with child-friendly definitions and Hindi

STATIC ELECTRICITY

An invisible force that builds up on objects when they are rubbed together. "Static" = staying in one place.

Hindi: Sthir vidyut (Sthir vidyut)

Eg: Kanghi attracting hair after rubbing

CHARGE

The invisible property of tiny particles that creates pulling or pushing forces between objects.

Eg: Too many charges = static electricity!

ATTRACT

To pull toward. A charged object attracts very light uncharged objects near it.

Hindi: Aakarshan

Eg: Charged comb attracts murmura

REPEL

To push away. Two objects with the same charge push each other apart.

Hindi: Praticarsha

Eg: Two charged balloons push away from each other

DISCHARGE

When built-up static charge is released suddenly — like a tiny spark or shock.

Eg: Touching a metal door handle — ZAP!

LIGHTNING

A massive discharge of static electricity between storm clouds or between clouds and the ground.

Hindi: Bijli (Bijli)

Eg: Monsoon storms over Indian cities

FORCE

A push or pull that can move objects. Static electricity is an invisible pulling force.

Hindi: Bal (Bal)

Eg: Charged comb creates a pulling force

PREDICT

Your best guess about what will happen BEFORE you test it — a key science skill!

Eg: I predict the murmura will jump toward the comb

OBSERVE

To look carefully and notice what is happening during the experiment.

Eg: I observed the murmura leaped 3 cm toward the comb

6. COMMON MISCONCEPTIONS

Watch for these in learner responses — and how to address them effectively

Misconception	Scientific Reality	Teaching Strategy
"Static electricity and magnetism are the same thing."	Both are invisible forces but completely different phenomena with different causes.	Show: "A magnet attracts iron and steel. A charged comb attracts murmura — not iron. Same shape of force, completely different science."
"The comb attracts murmura even without rubbing — any comb would do this."	The comb ONLY works AFTER being rubbed (charged). An uncharged comb does nothing.	Always demo the control test first: hold uncharged comb → nothing. Now rub it → murmura dances! Let the contrast speak.
"Static electricity is dangerous — like getting electrocuted from a socket."	Static charge in this experiment is tiny and completely harmless.	"The static in our experiment is much, much weaker than a socket. If you feel a tiny tingle, that is just the charge — and it is exciting, not dangerous!"
"The charged comb will attract everything near it — including heavy objects."	Static force is weak — it can only move very light, uncharged objects.	The stone/coin test makes this perfectly clear. Ask: "Is it heavier? Or does it have less charge?" Guide toward: weight determines whether static force is strong enough.
"Static electricity stays on the comb forever."	Charge dissipates over time — especially in humid conditions.	"Charge is like a soap bubble — it does not last! That is why we rub and then test quickly. On a humid monsoon day, it leaks away even faster!"
"Lightning is caused because clouds are wet with rain water."	Lightning is caused by charge separation within storm clouds — a massive static event.	Keep it simple: "Clouds rub together and build up charge. When it gets big enough — CRACK! Just like our kanghi, but a trillion times bigger!"

7. ASSESSMENT RUBRIC

5 criteria on a 4-point scale — for educator evaluation

Criteria	Excellent (4)	Good (3)	Developing (2)	Needs Support (1)
Concept Understanding	Clearly explains static electricity, rubbing, and charge using correct terms	Explains with minor gaps or errors	Partial explanation — some confusion about cause	Cannot yet explain the concept

Criteria	Excellent (4)	Good (3)	Developing (2)	Needs Support (1)
Observation Skills	Accurately describes which objects jumped/did not jump with detail on all 4 test materials	Describes most observations correctly (3 of 4)	Notes some movement but lacks specific detail	Cannot identify or describe observations
Control Test Understanding	Explains WHY the uncharged comb does nothing — links to rubbing creating charge	Notes uncharged comb does not work; partial explanation	Observes difference but cannot explain why	Did not notice or cannot recall the control test
Real-life Connections	Names 2+ Indian examples and explains HOW static is involved in each	Names 1 Indian example with explanation	Names an example without explaining the connection	Cannot name a real-life example
Prediction & Worksheet	Clear prediction with reasoning; complete observation table; labelled drawing	Prediction present; observation table mostly complete	Prediction made but observation recording incomplete	No prediction made; worksheet largely incomplete

8. STUDENT OBSERVATION WORKSHEET

Printable — to be completed during the investigation phases

Name: _____

Date: _____

■ BEFORE THE EXPERIMENT — My Prediction

I think the charged comb WILL attract: _____ because it is _____.

I think it will NOT attract: _____ because it is _____.

Circle your choice:

■ Murmura ■ Paper bits ■ Salt grains ■ Stone ■ Coin

■ DURING THE EXPERIMENT — Observation Table

Test Object	Heavy or Light?	Did it Jump? YES / NO	Why I think so
Murmura (puffed rice)	Very Light	YES / NO (circle)	_____ —
Torn paper bits	Light	YES / NO (circle)	_____ —
Salt grains	Very Light	YES / NO (circle)	_____ —
Stone / pebble	Heavy	YES / NO (circle)	_____ —
Coin	Heavy	YES / NO (circle)	_____ —

■ THE CONTROL TEST — Uncharged vs. Charged Comb

What happened when I held the **UNCHARGED comb** (before rubbing) above the murmura?

What happened AFTER rubbing the comb on hair?

Why was rubbing important? _____

■ AFTER THE EXPERIMENT — What I Learned

Static electricity is:

The objects that jumped were: _____ because they are: _____

The objects that did NOT jump: _____ because they are: _____

One Indian example of static electricity I can think of from my own life:

⇒ ■ Draw It! — Charged Comb Attracting Murmura

(Draw the charged comb with arrows showing murmura jumping TOWARD it. Label: "Charged Comb", "Murmura", "Direction of Attraction Force".)

9. INDIAN CONTEXT REFERENCE BANK

All content must feel genuinely Indian — draw from this bank across all outputs

Category	Indian Items / Examples	Static Connection
Rubbing Objects	Plastic kanghi / comb, plastic scale (ruler), plastic pen/biro, plastic water bottle, rubber balloon	All build static when rubbed on dry hair, wool, or cotton fabric
Light Attracted Objects	Murmura, namak (salt), chai patti (tea leaves), akhbaar bits (newspaper), thermocol from packaging, wool thread snippings	All light enough for weak static force to attract — ideal for the experiment
Rubbing Surfaces	Dry hair (preferred), woollen monkey cap / muffler, cotton school uniform sleeve, synthetic saree fabric, flannel cloth	Transfer electrons to plastic when rubbed; dry conditions essential
Clothing / Fabrics	Woollen sweater / cardigan (Delhi/Shimla winters), synthetic saree / dupatta, monkey cap, school uniform cotton	Wool and synthetic fabrics are excellent static generators due to surface texture
Seasonal Context	Dry winter (Oct–Feb): Best time — maximum static. Monsoon (Jun–Sep): Humidity reduces static — use only for lightning example. Dry summer interior India: Works well in dry regions.	CRITICAL: Warn educators that humid/rainy days significantly reduce experiment effectiveness
Daily Life Settings	Indian kitchen (thali, murmura bowl), Indian bathroom (mirror, dry hair), school classroom (plastic scale, notebook bits), local bazar (balloons), bus/school gate (metal shock)	Every setting connects to a static electricity example learners already know
Cultural Moments	Diwali / birthday party (balloons on walls), monsoon storm (lightning), school morning assembly (combing hair before assembly)	Connect session to culturally familiar experiences for maximum resonance

10. SUCCESS CRITERIA

Three levels — Learner, Content Quality, and Production Quality

■■■ Learner Success

- ◆ Names static electricity in one simple sentence with correct meaning
- ◆ Correctly identifies which objects a charged comb attracts — and gives a reason
- ◆ Explains why the UNCHARGED comb does nothing (the control test insight)
- ◆ Names at least one Indian real-life example of static electricity unprompted
- ◆ Completed both prediction and observation on worksheet (process over outcome)

- ◆ Scores 70%+ on the post-session quiz (5–6 questions)

■ Content Quality

- ◆ All 5 Learning Objectives addressed across the full output suite
- ◆ Minimum 3 authentic Indian examples in every major output
- ◆ The "uncharged comb control test" included in investigation AND in quiz
- ◆ Humidity / dry conditions caveat included in Teacher Guide and Parent Card
- ◆ Investigation materials are all locally available and low-cost
- ◆ Zero factual errors verified against NCERT and standard science references
- ◆ No premature introduction of "electron", "proton", or "atom" by name

■ Production Quality

- ◆ Learner engagement maintained throughout — visible excitement at murmura moment
- ◆ The "murmura jumping" animation is visually dramatic — large, clear, with spark effects
- ◆ Indian character illustrations — diverse skin tones, school uniforms or home wear
- ◆ All slides use minimum 28pt body text; all print text minimum 14pt
- ◆ Video narration at 110–130 words per minute (tested with timer)
- ◆ Worksheet printable in black-and-white with full readability
- ◆ Planrs Academy branding on all outputs; session title on all materials

Planrs Academy · Science Lab · Static Electricity — Dancing Cereal · Session 1

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"Learning that feels like play — for every Indian child."